evolution of
low-mass stars

Rubric for project presentations:

Content: 70%

- clearly stated goal of project: 15%
- description of methods to achieve goal (what worked and what didn’t?): 20%
- statement of results: 20%
- statement of future work that could be done: 15%

Presentation: 20%

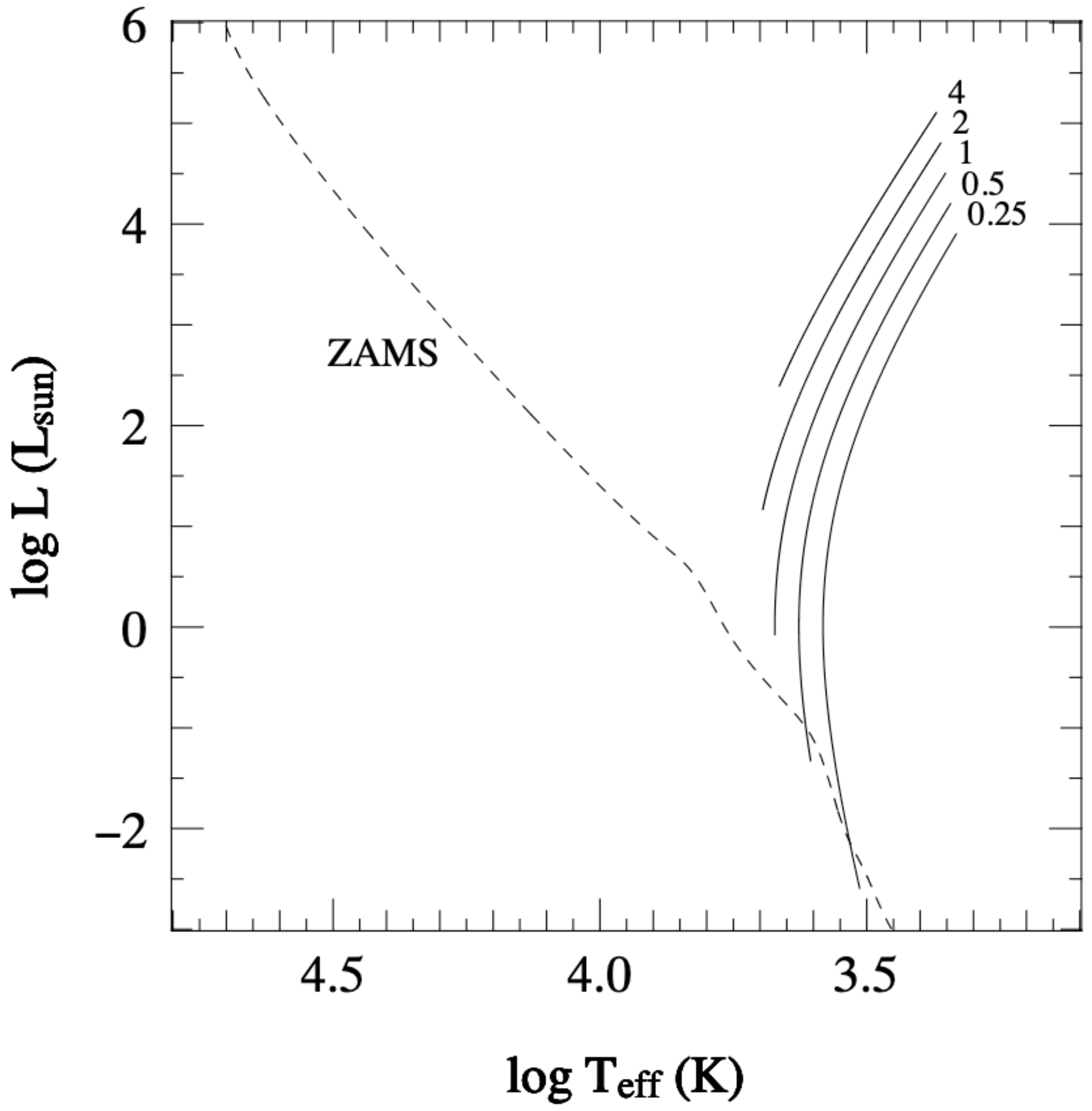
- correct presentation timing (10 mins): 10%
- thoughtful slides: 10%

Questions: 10%

- Asked at least one question during presentations

How are we all feeling about this?

Pre-main-sequence contraction



$$\tau_{PreMS} \simeq 10^7\,\mathrm{yr}\left(\frac{M}{M_\odot}\right)^{-2.5}$$

When stars are contracting onto the Zero-Age Main Sequence, we assume that they do so adiabatically because they are fully convective — this is called the Hyashi track

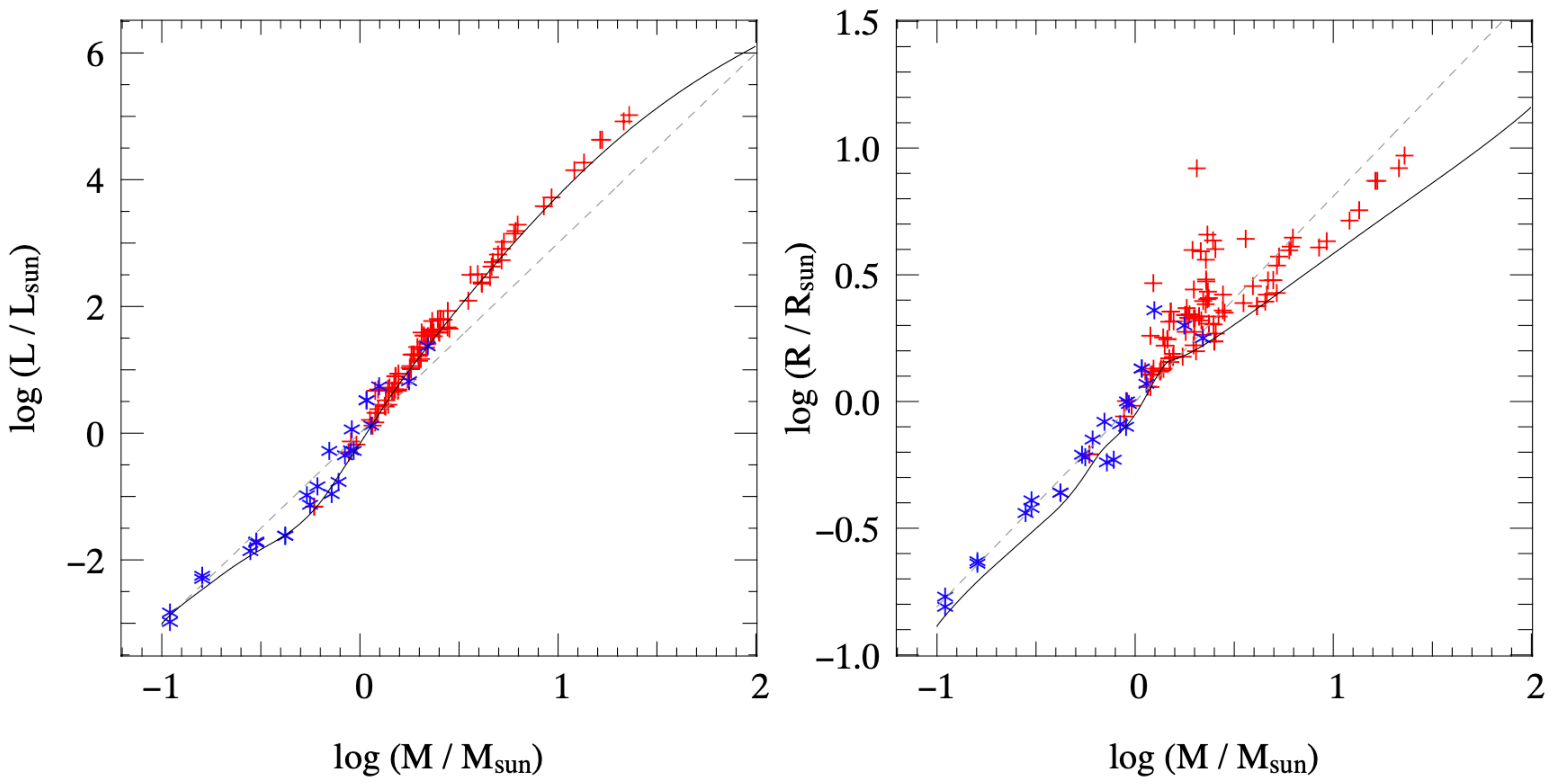
You can write down a heuristic argument for how this works following Section 9.1.1 in Onno's notes

The important things to take away are:

(1) more massive stars have higher temps as they contract

(2) more massive stars contract faster than less massive stars

ZAMS: only the metallicity and initial mass are needed to evolve



$$\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho}$$

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} - \frac{1}{4\pi r^2}\frac{\partial^2 r}{\partial t}$$

$$\frac{\partial l}{\partial m} = \epsilon_{\rm nuc} - \epsilon_\nu - T\frac{\partial s}{\partial t}$$

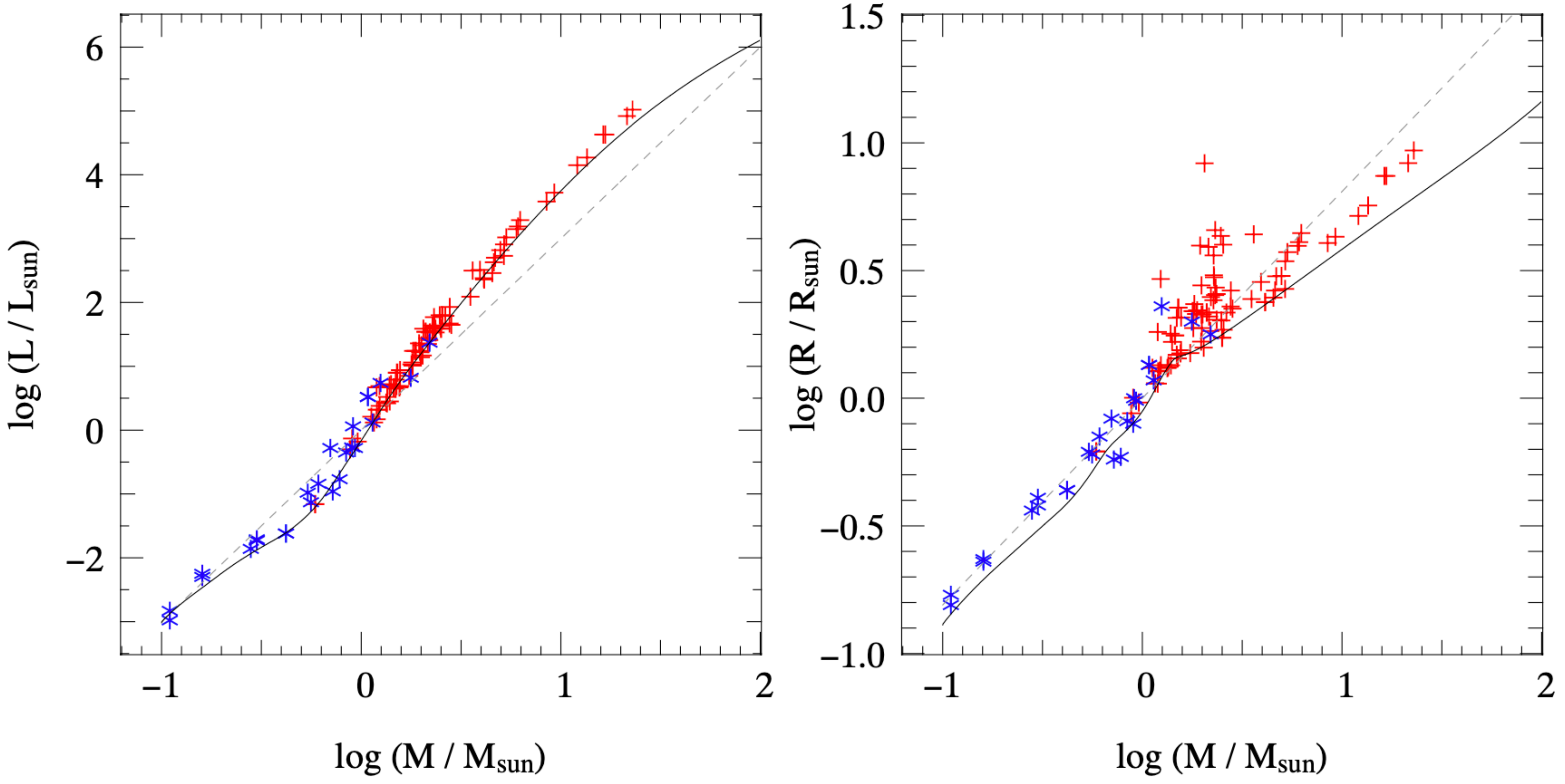
$$\frac{\partial T}{\partial m} = -\frac{Gm}{4\pi r^4}\frac{T}{P}\nabla \qquad \text{with} \qquad \nabla = \begin{cases} \nabla_{\rm rad} = \dfrac{3\kappa}{16\pi acG}\dfrac{lP}{mT^4} & \text{if } \nabla_{\rm rad} \le \nabla_{\rm ad} \\ \nabla_{\rm ad} + \Delta\nabla & \text{if } \nabla_{\rm rad} > \nabla_{\rm ad} \end{cases}$$

$$\frac{\partial X_i}{\partial t} = \frac{A_i m_u}{\rho}\left(-\sum_j (1+\delta_{ij})\, r_{ij} + \sum_{k,l} r_{kl,i}\right) \quad [\text{+ mixing terms}] \qquad i = 1 \ldots N$$

Different colors are different data sets containing eclipsing binaries

Dotted lines: polytrope for CNO cycle   Solid lines: stellar evolution code

ZAMS: only the metallicity and initial mass are needed to evolve

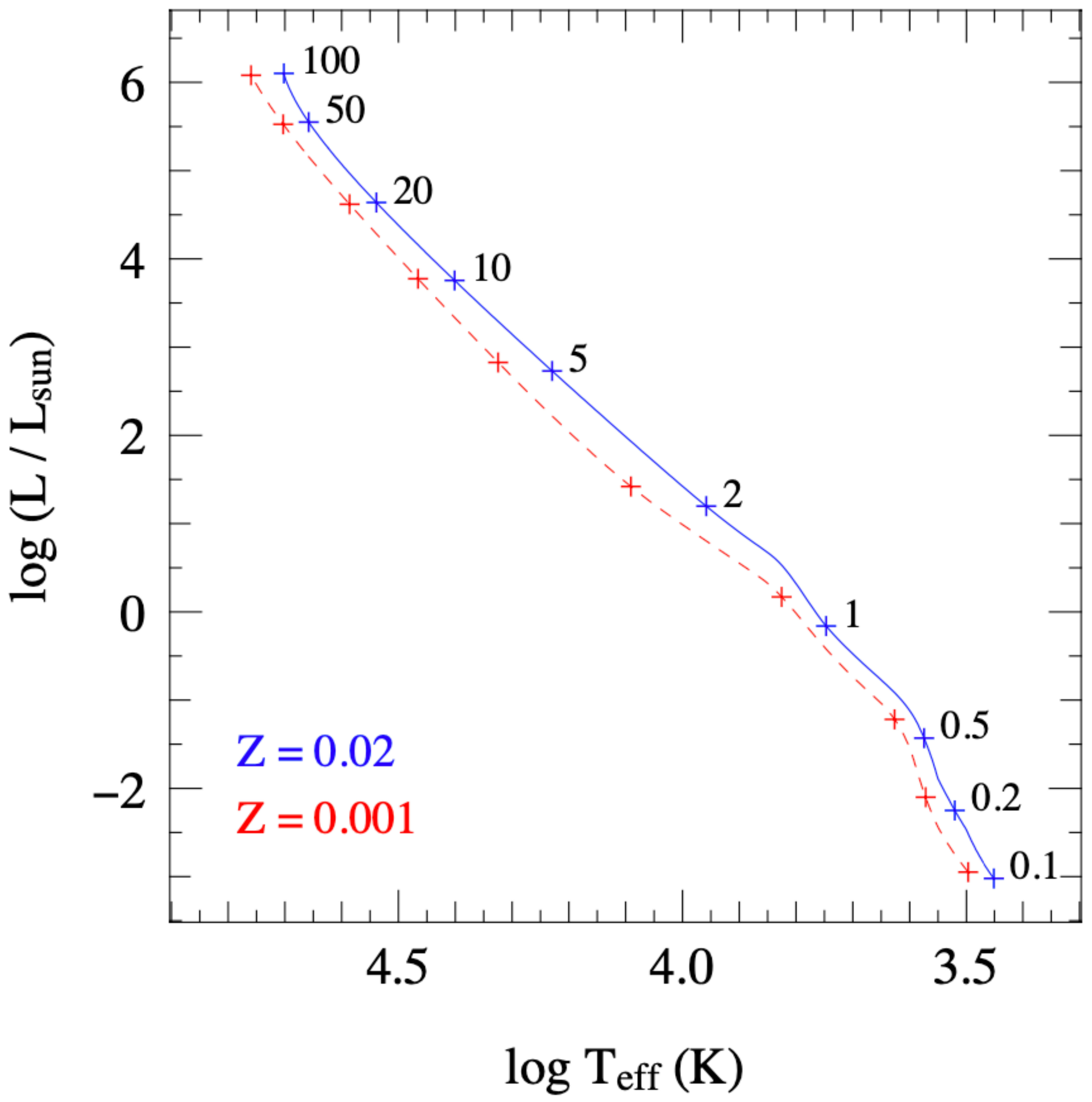


L stays relatively constant on MS

R does not — especially for more massive stars

Different colors are different data sets containing eclipsing binaries

Dotted lines: polytrope for CNO cycle Solid lines: stellar evolution code

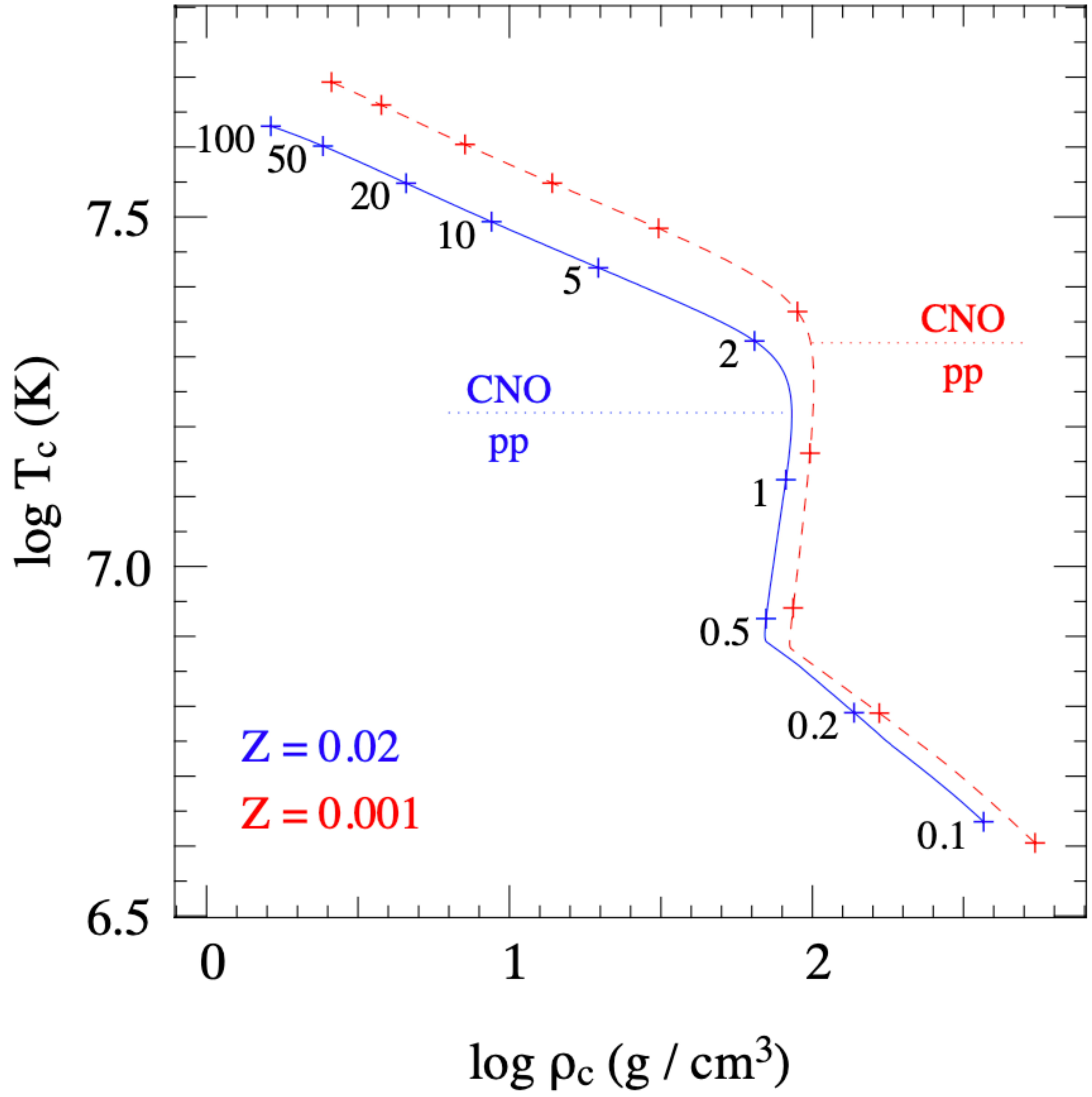


Steepness of L with T is due to shallow M—R relation and steep M—L relation

Lower metallicities leads to:

- Smaller radii
- Hotter temperatures
- Brighter luminosity (below 5 $M_\odot$)

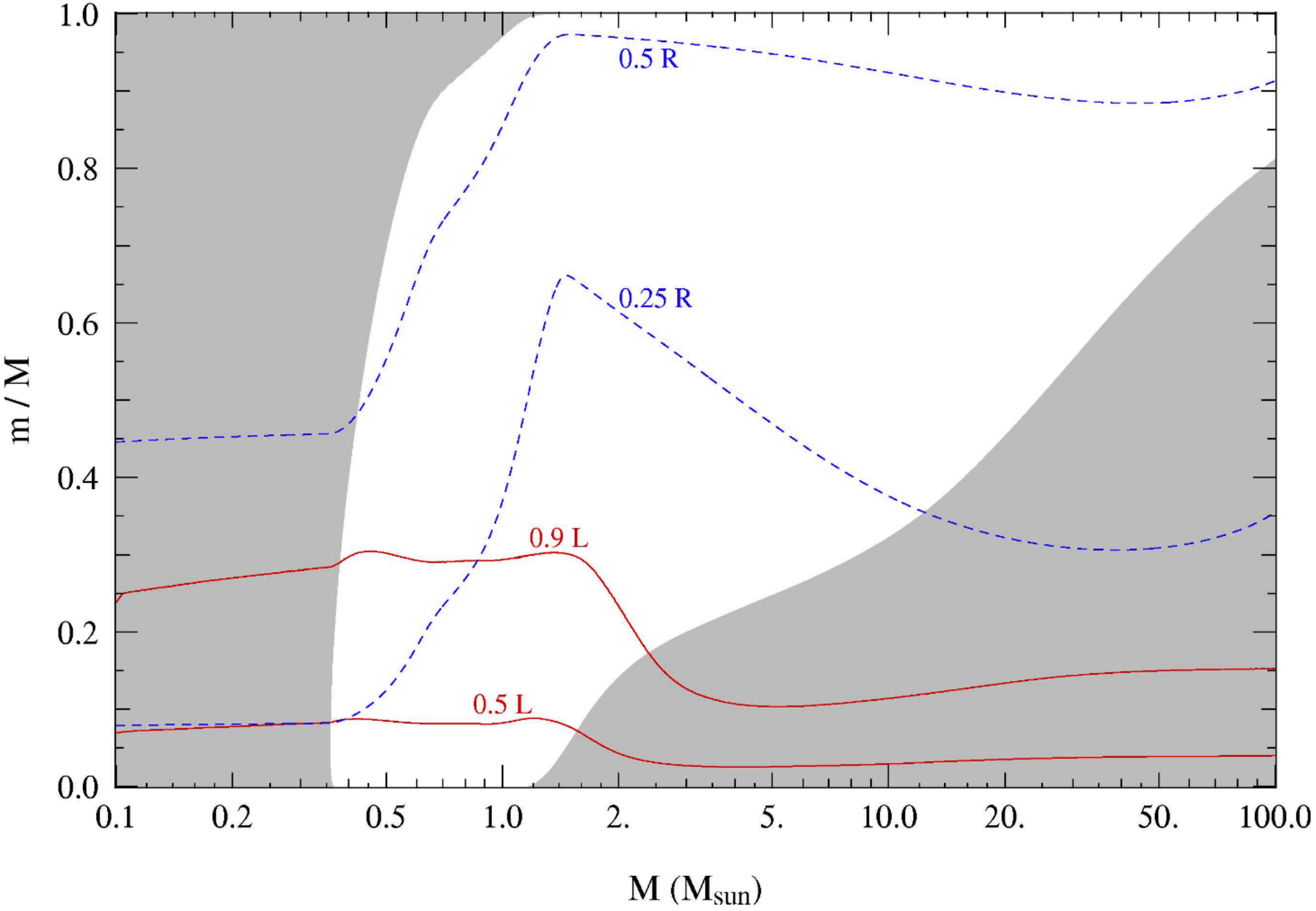
Due to lower bound-free opacity at lower metallicity as well as central conditions like temperature and density



CNO cycle vs pp chain enters at different temperatures for different Z due to differences in CNO abundances

$$\bar{T} = \frac{\alpha}{3} \frac{\mu m_{\mathrm{u}}}{k} \frac{GM}{R}$$

Same mass, same luminosity, higher temp: smaller radii



Very low mass stars are fully convective

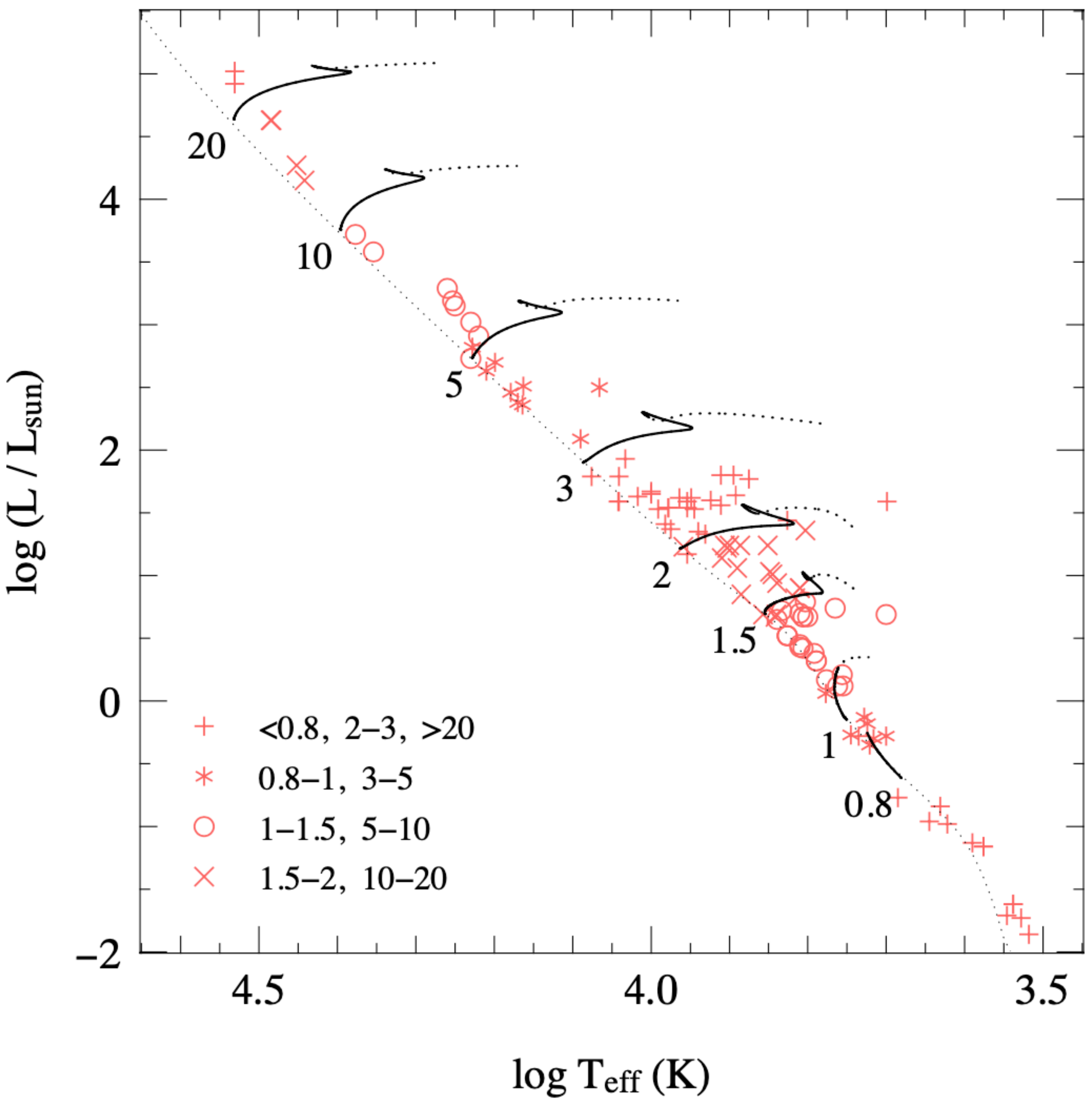
Sun-like stars have radiative cores and convective envelopes

$\nabla_{rad} < \nabla_{ad}$

Massive stars have convective cores and radiative envelopes

$\nabla_{rad} > \nabla_{ad}$

Central hydrogen burning:
both HE and TE are maintained; evolution occurs due to chemical evolution only



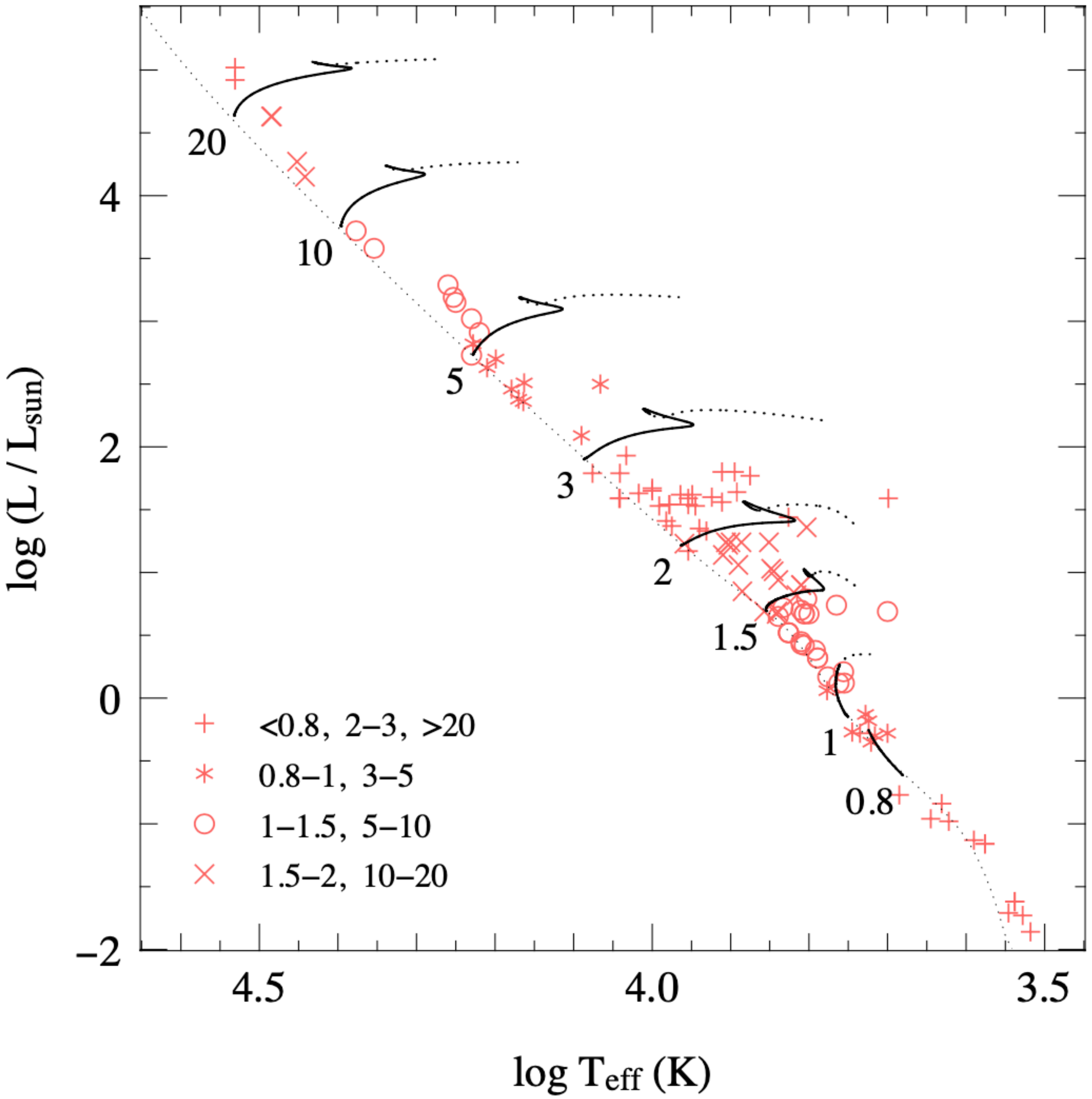
Low mass stars evolve toward hotter temps and have modest radial growth

Higher-mass stars have strong radial growth and evolve toward lower temps

Chemical evolution results in:

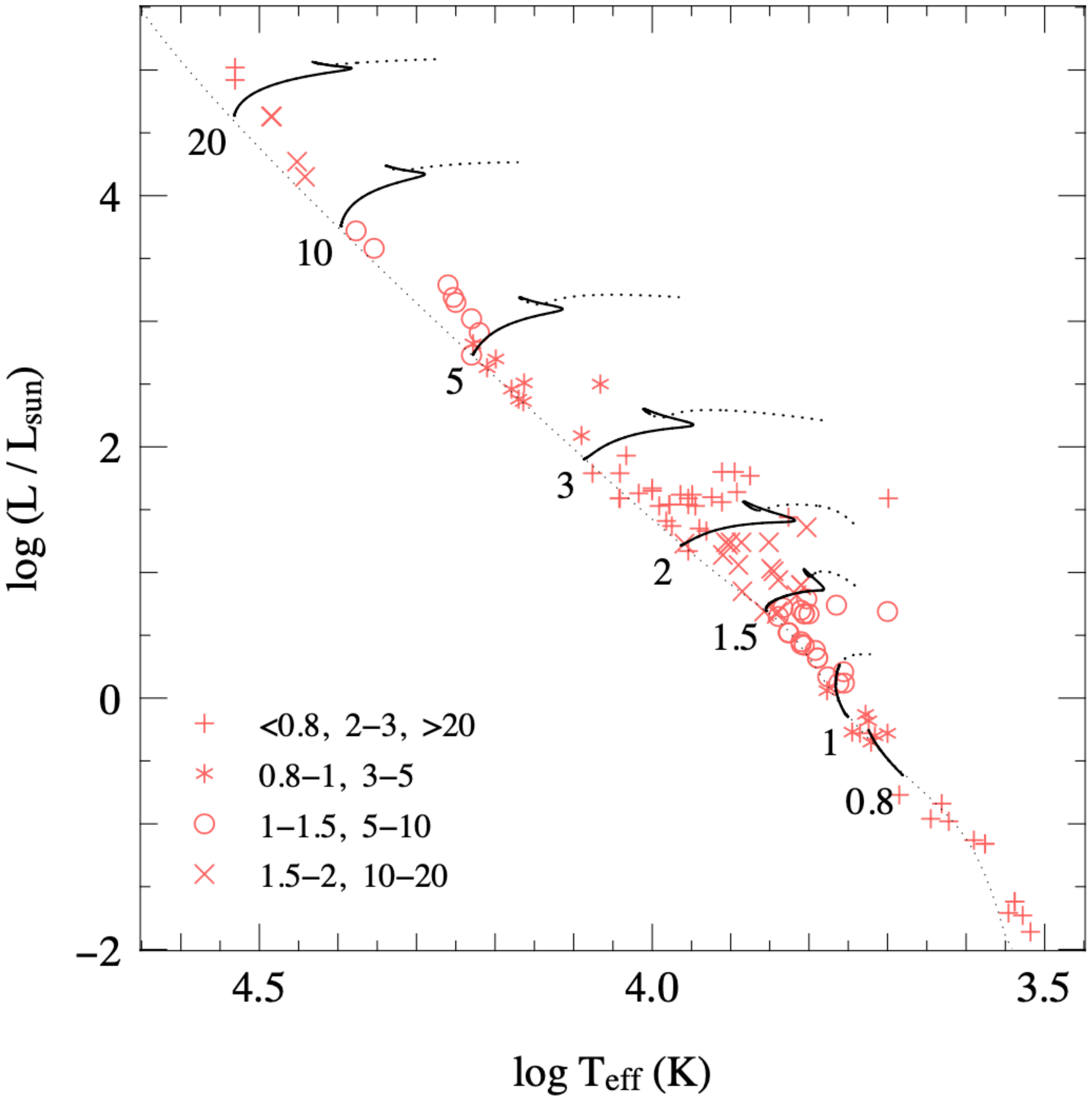
- Increased μ, which increases L but lowers Pressure
- T_{central} remains constant because nuclear reactions act like a thermostat ($\epsilon_{\mathrm{pp}} \propto T^4, \epsilon_{\mathrm{CNO}} \propto T^{18}$)

These lead to core-envelope structures



CNO-powered evolution:

- convective core allows HE and TE to occur since the nuclear production rate (and thus T) stays constant; convective core also mixes the composition and thus increases fuel available
- Near end of MS, core H becomes depleted and the core T must increase
- Once H is fully exhausted, the star is losing more energy at surface than it is producing and must contract until CNO cycle is reignited in a shell around the He core (H-shell burning)



pp-powered evolution:

- Smaller T dependence from ϵ_{pp} means that core T can increase more than in CNO
  - This means less outer-layer expansion
- Lower T dependence also leads to radiative core, so dX/dt is peaked in core and has a smooth transition to hydrogen shell burning
  - No hook feature
- M=1.1-1.3$M_\odot$ transition from pp to CNO near end of MS and thus develop convective cores

$$\frac{\mathrm{d}X}{\mathrm{d}t} = -\frac{\epsilon_{\mathrm{nuc}}}{q_{\mathrm{H}}}.$$

Integrate over all mass shells

$$\frac{\mathrm{d}M_{\mathrm{H}}}{\mathrm{d}t} = -\frac{L}{q_{\mathrm{H}}},$$

Integrate over the MS lifetime

$$\Delta M_{\mathrm{H}} = \frac{1}{q_{\mathrm{H}}} \int_0^{\tau_{\mathrm{MS}}} L\, \mathrm{d}t = \frac{\langle L \rangle \tau_{\mathrm{MS}}}{q_{\mathrm{H}}},$$

<L> is average luminosity over MS

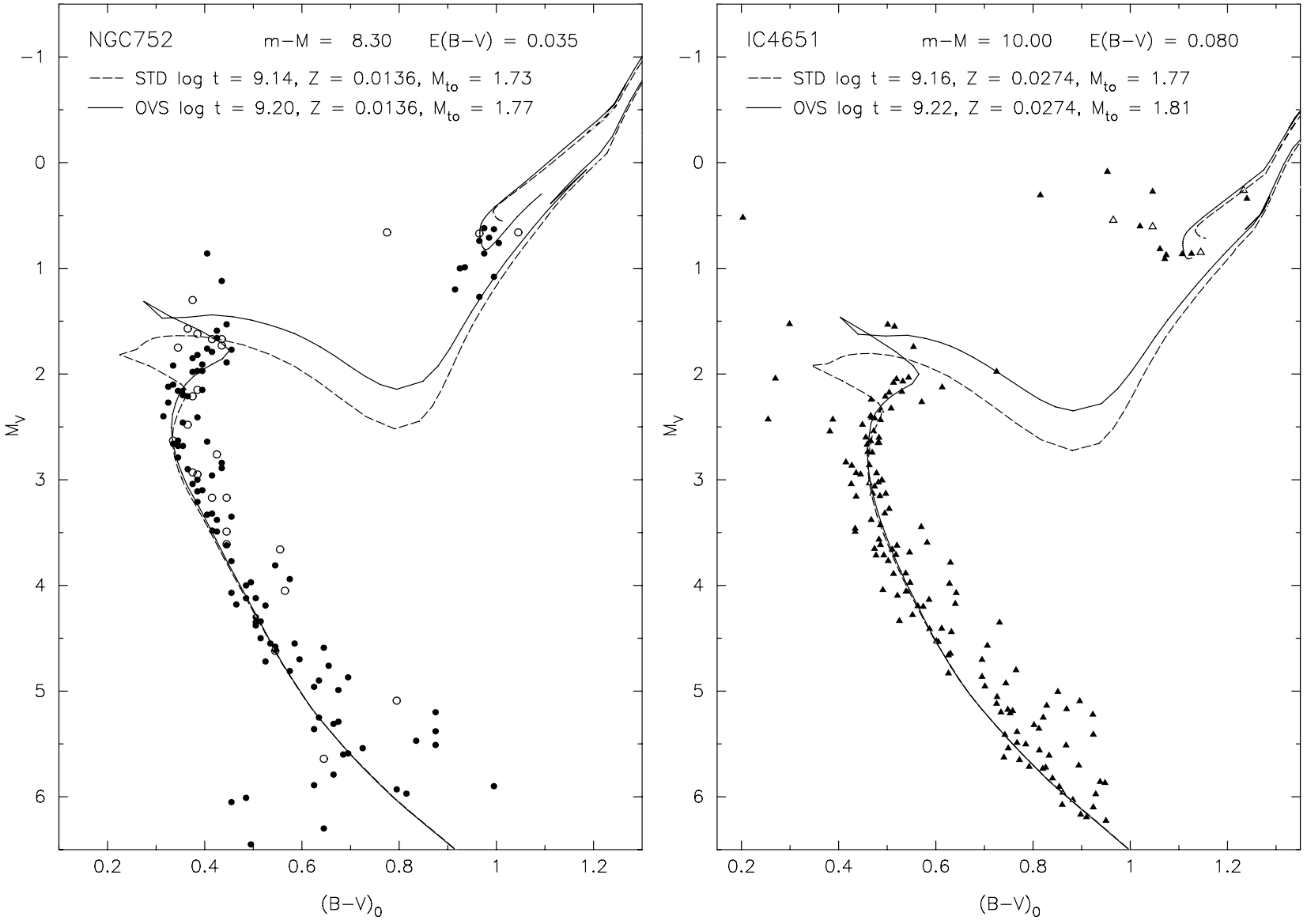
$$\Delta M_{\mathrm{H}} = f_{\mathrm{nuc}} M \qquad f_{\mathrm{nuc}} = X_0 q_c$$

X_0 is initial H mass fraction and q_c is core mass fraction at H depletion

$$\tau_{\mathrm{MS}} = X_0\, q_{\mathrm{H}} \frac{q_{\mathrm{c}} M}{\langle L \rangle}.$$

<L> is average luminosity over MS;
L—M relation shows that τ_{MS} decreases strongly for massive stars

Convective overshooting



More convection leads to

- Longer MS lifetime because mixing brings H to core
- Larger L,R increases due to μ increases
- Larger H-exhausted cores and thus brighter & shorter post-MS evolution

This is a knob to turn in stellar evolution codes

Post-main-sequence evolution

low-mass stars are those that develop a degenerate helium core after the main sequence, leading to a relatively long-lived *red giant branch* phase. The ignition of He is unstable and occurs in a so-called *helium flash*. This occurs for masses between 0.8 $M_\odot$ and $\approx$ 2 $M_\odot$ (this upper limit is sometimes denoted as M_{HeF}).

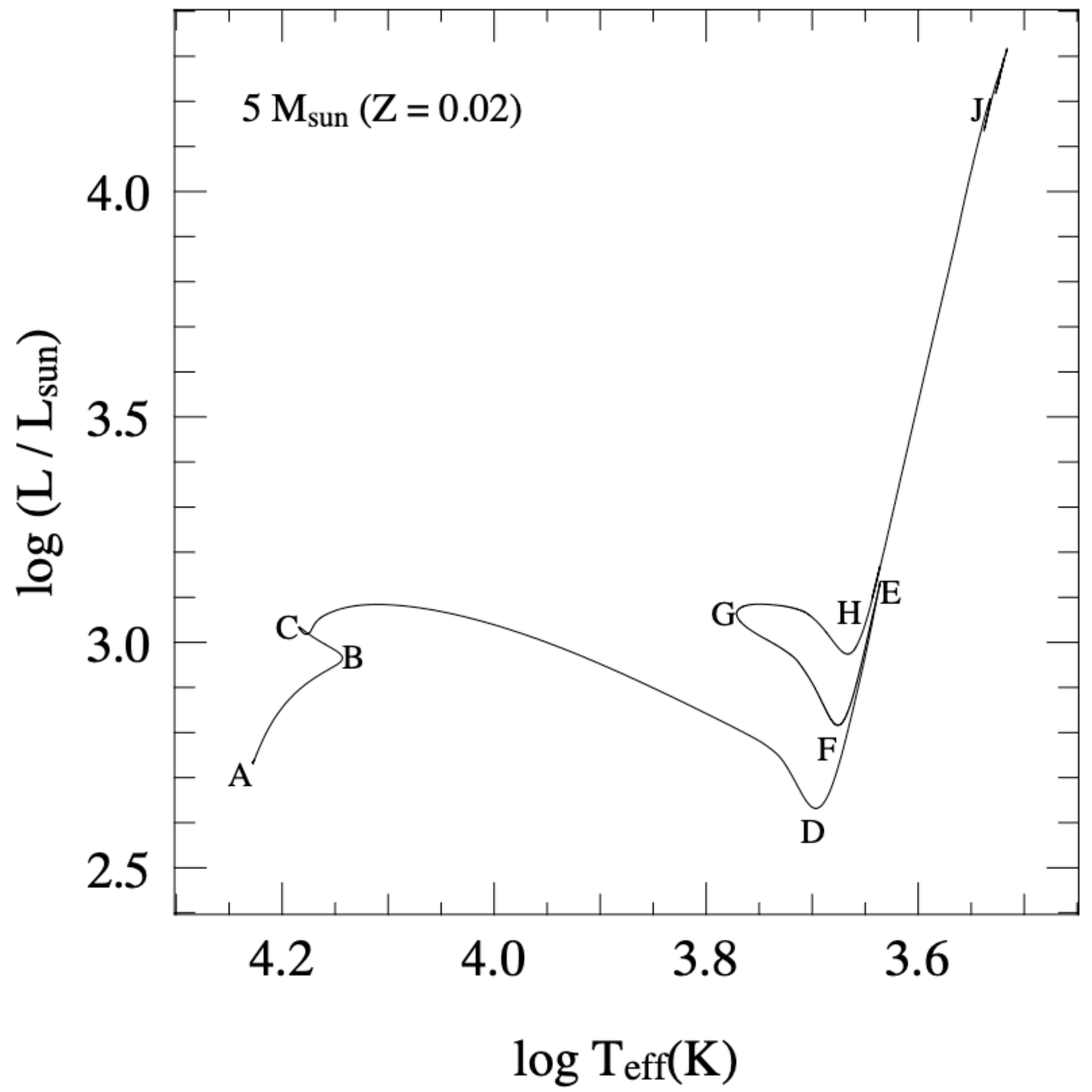
intermediate-mass stars develop a helium core that remains non-degenerate, and they ignite helium in a stable manner. After the central He burning phase they form a carbon-oxygen core that becomes degenerate. Intermediate-mass stars have masses between M_{HeF} and $M_{\mathrm{up}} \approx 8\,M_\odot$. Both low-mass and intermediate-mass stars shed their envelopes by a strong stellar wind at the end of their evolution and their remnants are CO white dwarfs.

massive stars have masses larger than $M_{\mathrm{up}} \approx 8\,M_\odot$ and ignite carbon in a non-degenerate core. Except for a small mass range ($\approx 8 - 11\,M_\odot$) these stars also ignite heavier elements in the core until an Fe core is formed which collapses.

Post-MS evolution

Whenever a star has an *active shell-burning source*, the burning shell acts as a *mirror* between the core and the envelope:

core contraction	$\Rightarrow$	envelope expansion
core expansion	$\Rightarrow$	envelope contraction



A: ZAMS

B: Contraction due to H depletion

C: Begin H-shell burning

C-D: thick shell burning, 2e6 yr total, 1e5yr from logT~4-3.7

D-E: Red giant w/ deep convective envelope, dredge up

E: Helium ignition

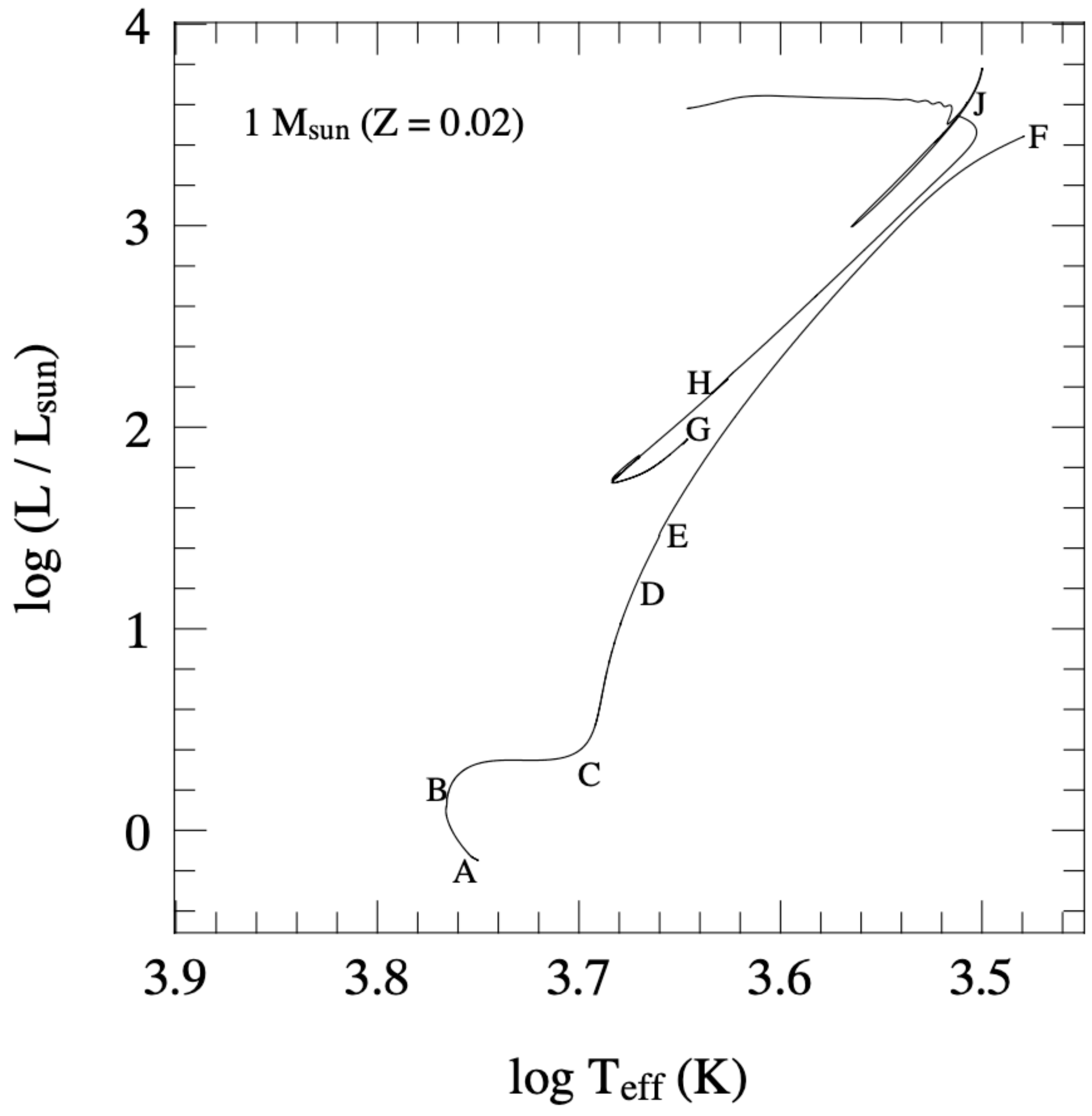
F: Envelope has become radiative

G: blue loop

H: end of core Helium burning

J: Asymptotic giant branch

Post-MS evolution



$$L \approx 2.3 \times 10^5 L_\odot \left(\frac{M_c}{M_\odot}\right)^6$$

$$M_c \lesssim 0.5\, M_\odot$$

A: ZAMS

B: near H depletion at 9 Gyr

B-C: Core grows in mass and contracts; "subgiants"

C: Core is degenerate and envelope is convective: base of GB

D: Convective envelope reaches layers processed by H-burning; —>first dredge up

E: H shell is beyond discontinuity of core-envelope; burning slightly decreases and you see a pileup of stars there

F: Helium flash: helium burning is unstable and results in thermonuclear runaway because temperature increases rather than decreases; neutrino losses help with carrying away heat

G: stable core He burning, H-shell burning.

H: end of core Helium burning

J: Asymptotic giant branch

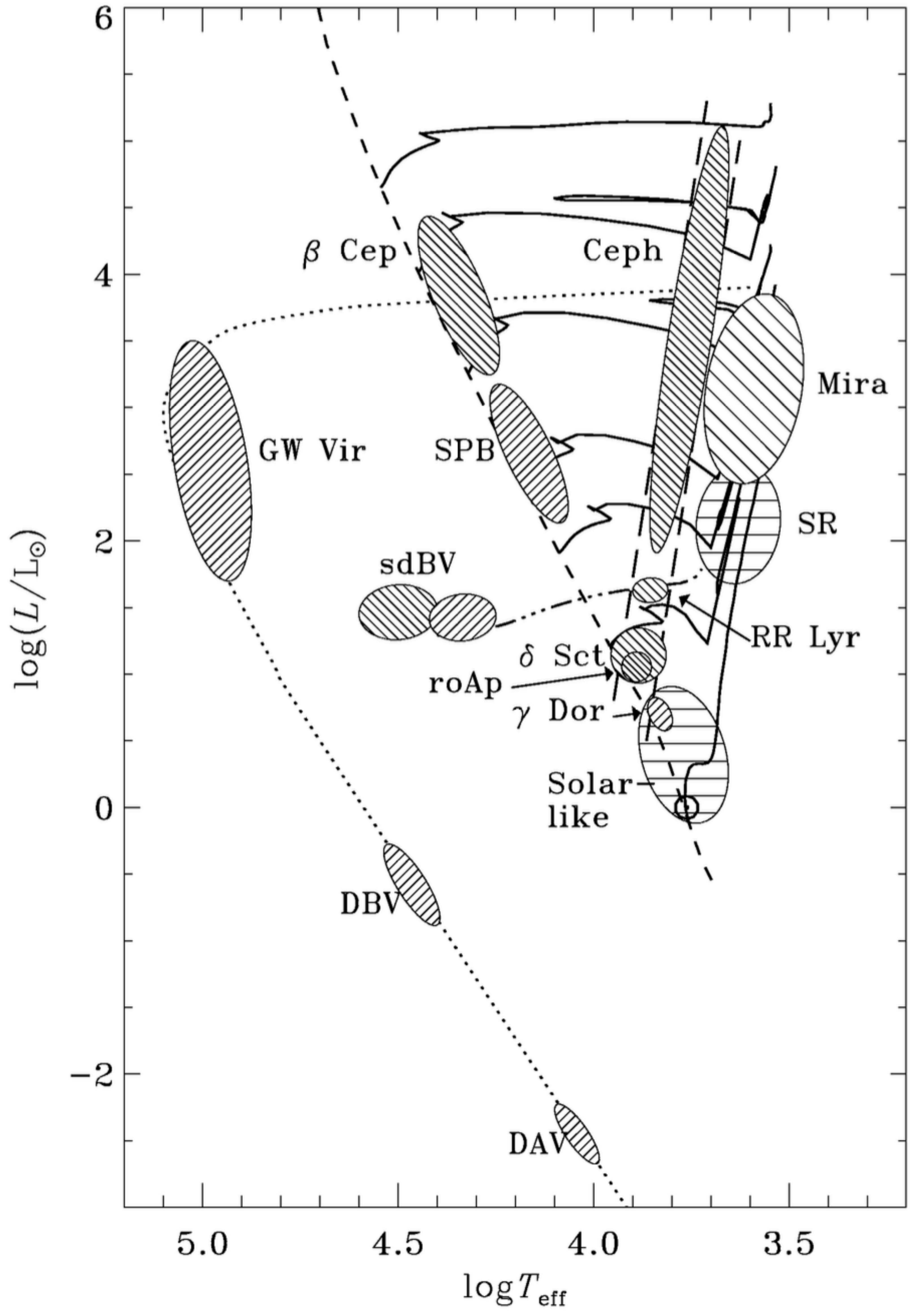


Figure 10.11. Occurrence of various classes of pulsating stars in the H-R diagram, overlaid on stellar evolution tracks (solid lines). Cepheid variables are indicated with 'Ceph', they lie within the pulsational instability strip in the HRD (long-dashed lines). Their equivalents are the RR Lyrae variables among HB stars (the horizontal branch is shown as a dash-dotted line), and the δ Scuti stars (δ Sct) among main-sequence stars. Pulsational instability is also found among luminous red giants (Mira variables), among massive main-sequence stars – β Cep variables and slowly pulsating B (SPB) stars, among extreme HB stars known as subdwarf B stars (sdBV) and among white dwarfs. Figure from Christensen-Dalsgaard (2004).

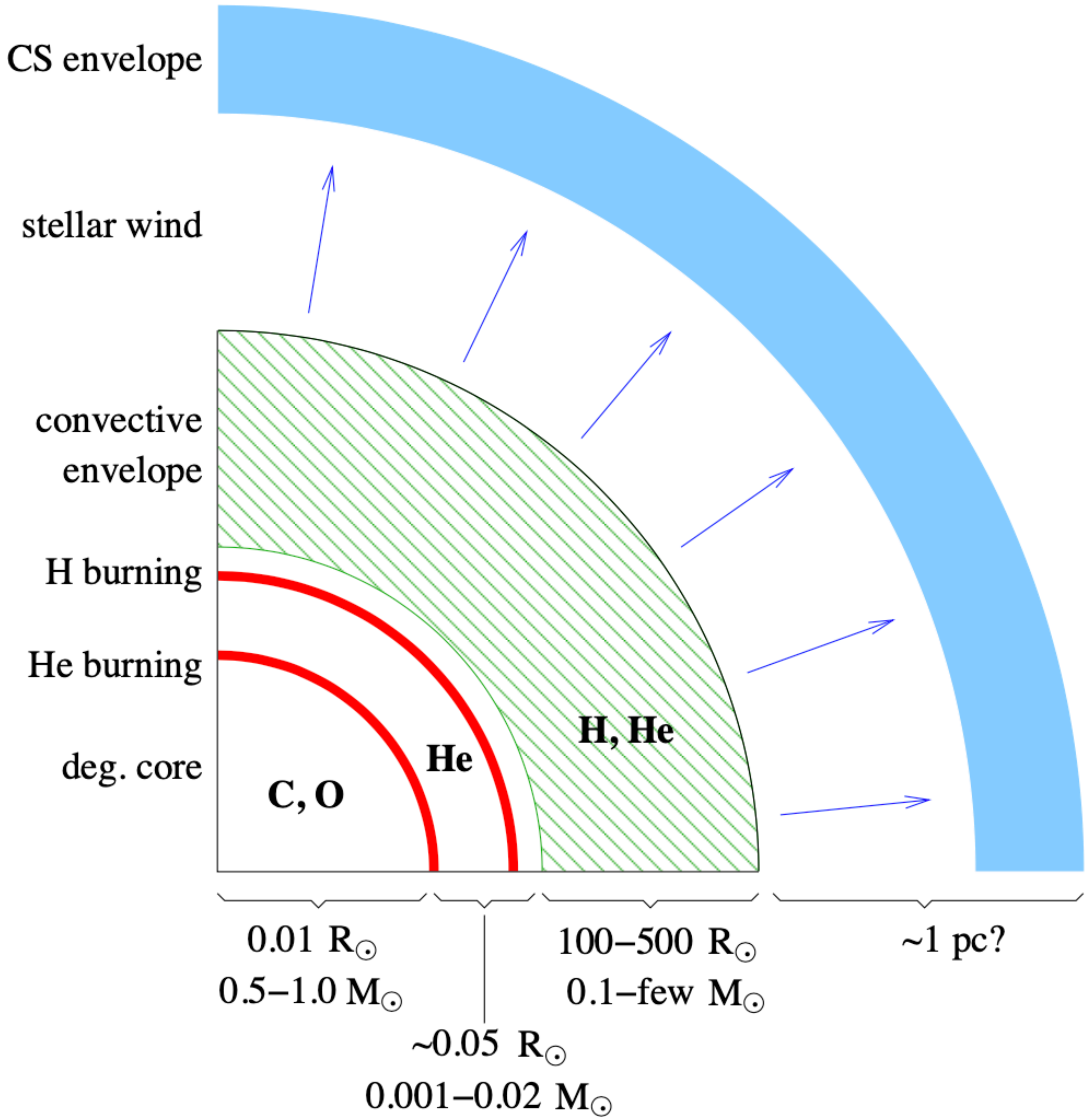
Pulsations can be seen as periodic photometric variability and can be used to test different stellar models as well as in things like the distance ladder for Cepheid variables whose pulsations depend on their luminosity and thus you have an independent distance measure

Wind mass loss occurs on the giant branch for low & intermediate mass stars

$$\dot{M} = -4 \times 10^{-13} \eta \frac{L}{L_\odot} \frac{R}{R_\odot} \frac{M_\odot}{M} M_\odot/\mathrm{yr}$$

$$\eta \sim 0.25 - 0.5$$

$1M_\odot$ star loses $0.3\,M_\odot$ of envelope mass by tip of giant branch



2nd phase of dredge up for intermediate mass stars leads to shells of H and He burning that occur at different rates

Thermally unstable He burning (like tiny helium flashes) leads to thermal pulses which can lead to nucleosynthesis of C, N, and > Fe elements (s-process)

$$L = 5.9 \times 10^4 L_\odot \left(\frac{M_c}{M_\odot} - 0.52 \right),$$

Thermally pulsing AGB stars can remove the entire envelope of the star leaving behind a planetary nebula and white dwarf in the core

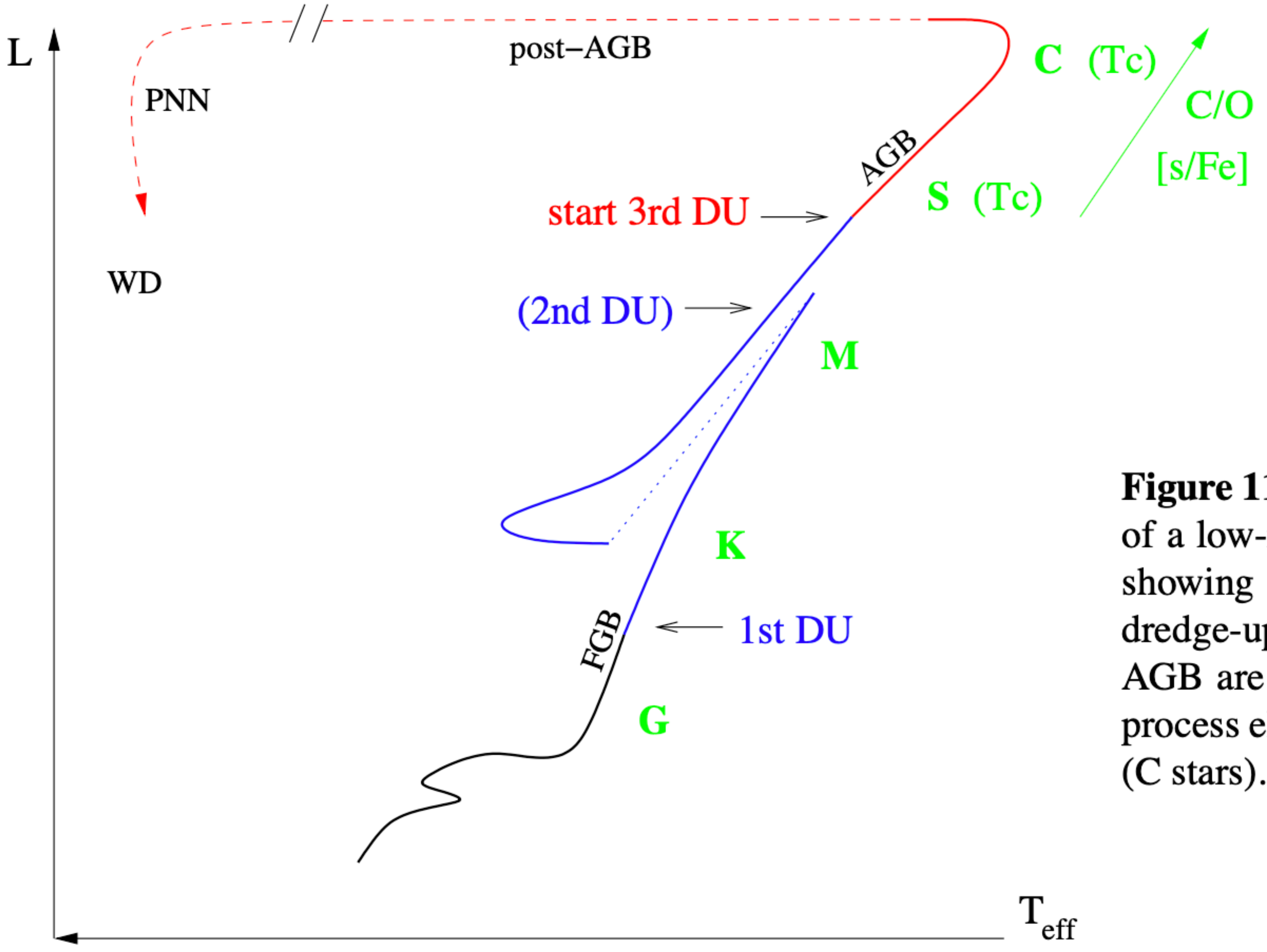


Figure 11.5. Schematic evolution track of a low-mass star in the H-R diagram, showing the occurrence of the various dredge-up episodes. Stars on the upper AGB are observed to be enriched in s-process elements (S stars) and in carbon (C stars).